1 THE PROBLEM

A lunar base is days from a hospital, and launching splints in every size wastes mass (~$1,000 per gram). It needs a custom splint it can print on demand from a scan.

$1M	1/6 g	−1–2%
launch cost per kilogram	lunar gravity	bone density per month

LATERAL ANKLE LIGAMENT TEAR
Outer ankle · heals in 3–26 weeks · caused by uneven flooring

CRITERIA

- Comfortable
- One-handed
- Durable 6 mo.
- Safe
- Custom fit
- Lunar innovation

CONSTRAINTS

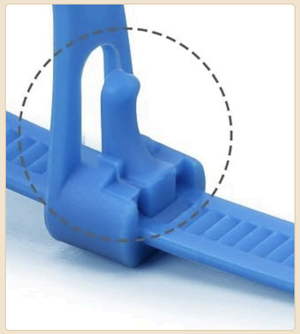
- 1/6 gravity
- Bone loss
- No hospital
- ~$1,000 / gram
- Printer limits
- Little space

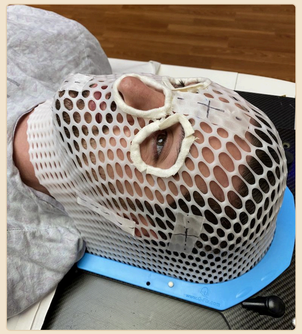
2 RESEARCH



SAM splint from our trainer


Splinted ankle, wrapped


Reusable zip tie: closure idea


Mesh mask molded to one person: scan-to-fit idea

> *"A splint allows almost no motion. Make it firm and tight."*
> School athletic trainer, 9/18



Survey of 56 students, Aug 2026

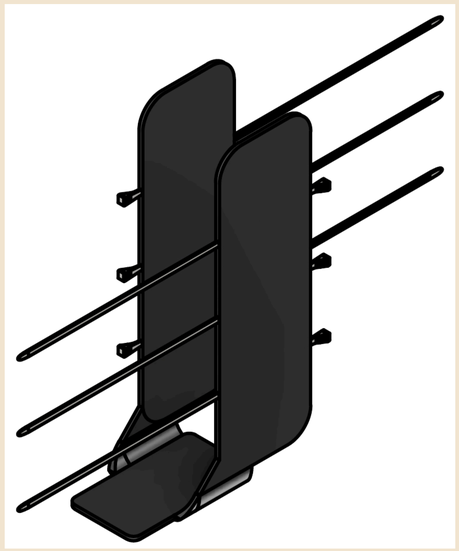
OUR DESIGN: PROS AND CONS

Pros
- Breathable
- Sanitary
- Adjustable

Cons
- Uses a lot of material

INNOVATION 1
SINGLE-PIECE RATCHET CLOSURE
Zip-tie straps, no loose parts, one hand.

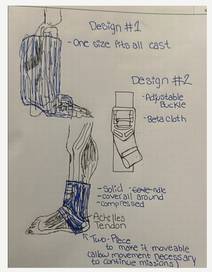
FEATURE
CUSTOM FIT
Printed from a scan of the leg.


Updated 3D CAD · Jose P. Hernandez, 10/16/2026

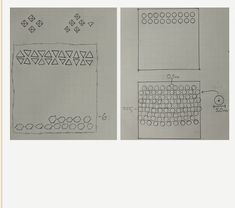
INNOVATION 2
INTEGRATED HYGIENE
Clean and check it over weeks of wear.

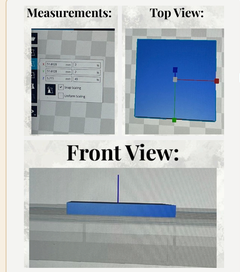
INNOVATION 3
THERMAL PHASE DESIGN
TPU flexible to fit, firm to hold.

3 FROM CONCEPT 1 TO OUR UPDATED DESIGN

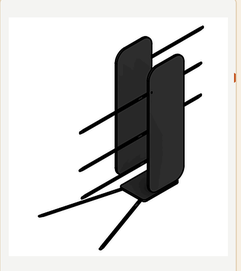


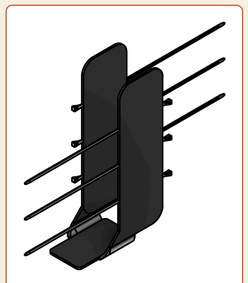
CONCEPTS 1 & 2
Cast, then buckle
A cast is not a splint, so we switched. Concept 2 added a buckle. · Nancy

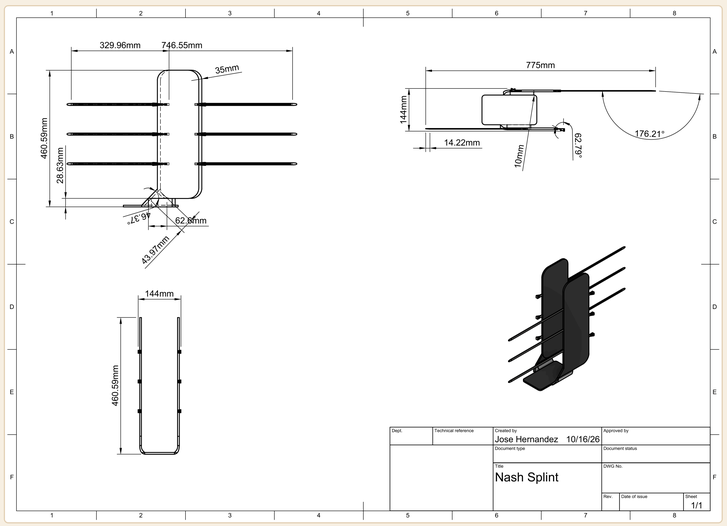

TEST
Hole patterns
Triangles failed, circles bent too much, so the wall went solid. · Gertrudis



CAD
Test samples
51.6 × 51.6 × 5.2 mm squares to compare materials.


BEFORE
Preliminary design
Our first CAD: latch + zip-tie straps in an X.


NOW
Updated design
We changed the CAD. New drawing, 10/16/2026.



Updated detail drawing with measurements (mm), Jose P. Hernandez, 10/16/2026

PARTS AND SIZES

Side plates (2)
35 mm rounded corners, 144 mm apart

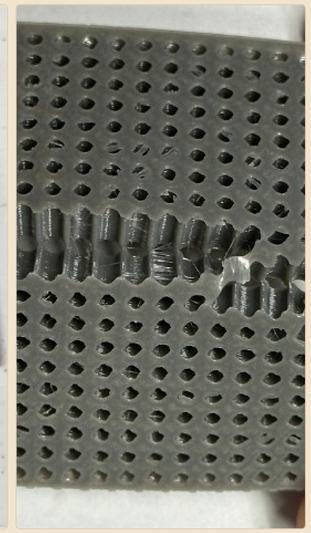
Base plate
Joins the two side plates at the bottom

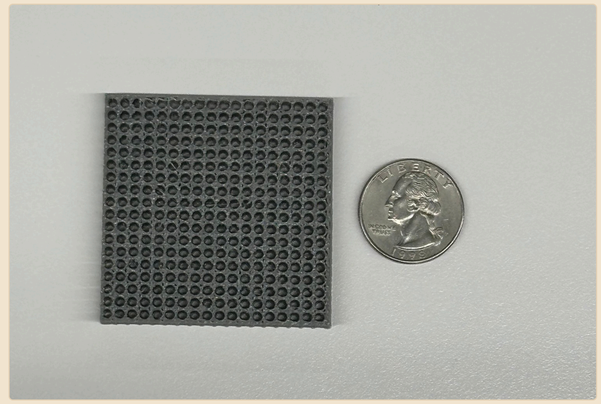
Straps
One-piece zip-tie style, X wrap

Overall
460.59 mm tall

3 TESTING


Hanging-weight test


Printed sample, split


TPU sample with quarter

PETG, holes: broke after bending	REJECTED
TPU, holes: layers split at 2,550 g	REJECTED
Triangle holes: did not hold	REJECTED
Solid TPU: firm, moulds slightly	CHOSEN
Straps at key points: ankle stayed still	CHOSEN

MATERIALS

PLA
Prototype 1

ABS
Prototype 2

TPU
Prototype 3 · flexes, firm when solid

~~PETG~~
Broke when bent

CF-PEEK
Goal flight material

NEXT STEPS

- →Strap thickness and break load
- →Filament softening temperature
- →Re-test small vents
- →Mass and print time from final CAD

Sept 24	Oct 21–22	Nov–Dec	Mid-April
Solid prototype	PDR, Mercedes	Prototype #2	NASA HUNCH evaluation

A NASA HUNCH 2026–27 student design project. Not endorsed by NASA. Not a medical device.